

# **JNTU Kakinada**

## **REGULAR EXAMINATION ? 2024**

Digital Logic Design ? Regular Examination, 2024

Subject Code: CS302 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Number Systems and Codes)**

1. Describe the working principle of Number Systems and Codes with necessary equations. [12 Marks]
2. Solve a problem related to Number Systems and Codes. Show all the steps clearly. [8 Marks]
3. List the advantages and limitations of Number Systems and Codes. Explain any two in detail. [12 Marks]

### **SECTION B (Unit 2: Boolean Algebra and K-Maps)**

4. Explain Boolean Algebra and K-Maps with a neat diagram/sketch and a suitable example. [8 Marks]
5. List the advantages and limitations of Boolean Algebra and K-Maps. Explain any two in detail. [12 Marks]
6. Write short notes on Boolean Algebra and K-Maps. [12 Marks]

### **SECTION C (Unit 3: Combinational Circuits)**

7. Explain Combinational Circuits with a neat diagram/sketch and a suitable example. [12 Marks]
8. Define Combinational Circuits. Explain its significance in Digital Logic Design with a suitable example. [8 Marks]
9. Compare and contrast the key concepts involved in Combinational Circuits. [12 Marks]

### **SECTION D (Unit 4: Sequential Circuits)**

10. Define Sequential Circuits. Explain its significance in Digital Logic Design with a suitable example. [12 Marks]
11. Derive the expression related to Sequential Circuits and explain each term. [12 Marks]
12. Write short notes on Sequential Circuits. [12 Marks]

### **SECTION E (Unit 5: Memory and Programmable Logic)**

13. Describe the working principle of Memory and Programmable Logic with necessary equations. [12 Marks]

**14. Solve a problem related to Memory and Programmable Logic. Show all the steps clearly. [12 Marks]**

15. Define Memory and Programmable Logic. Explain its significance in Digital Logic Design with a suitable example. [12 Marks]

\*\*\* End of Question Paper \*\*\*